

Feature Cross-Substitution in Adversarial Classification and Scalable Optimization of Randomized Operational Decisions in Adversarial Classification Settings

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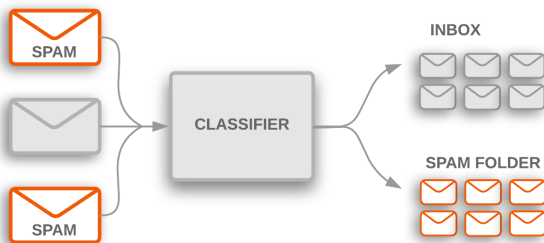
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Overview

- 1 Motivation
- 2 Feature Selection
- 3 Operational Decision

Machine Learning in security settings

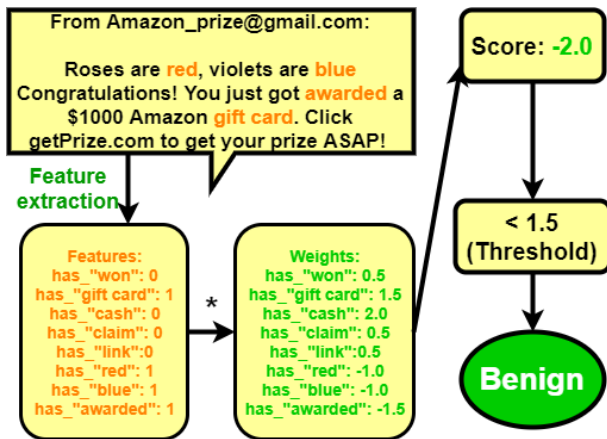
- Common use cases: fraud detection, computer intrusion detection, and web search
- Very important tool for filtering spam and phishing emails



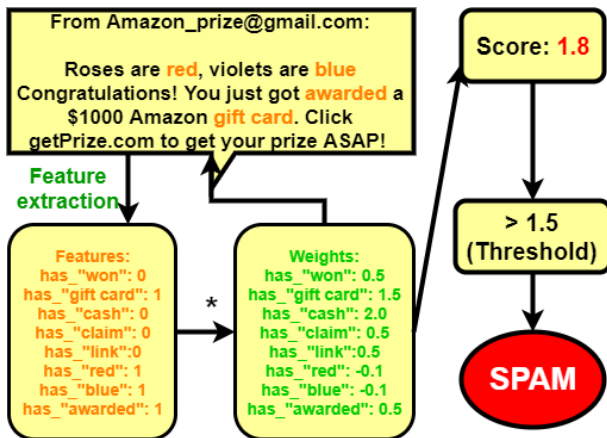
Example of spam classification



Many researches didn't consider adversarial modification

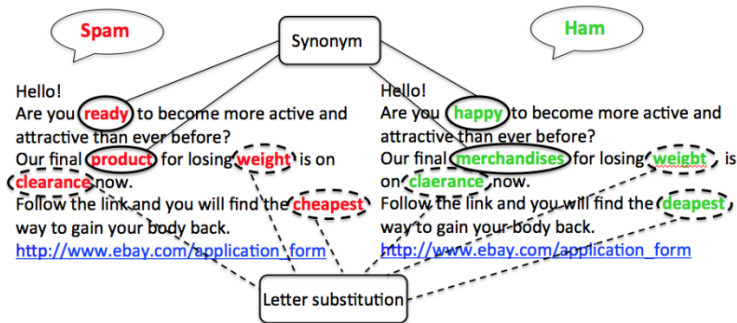


Needs to retrain to counteract



Feature Cross-Substitution

- Adversary substitutes words for their synonyms or substitutes similar-looking letters in words
- Can achieve a much lower cost of transforming



Need to redefine the cost function

- Traditional distance based cost(L1):

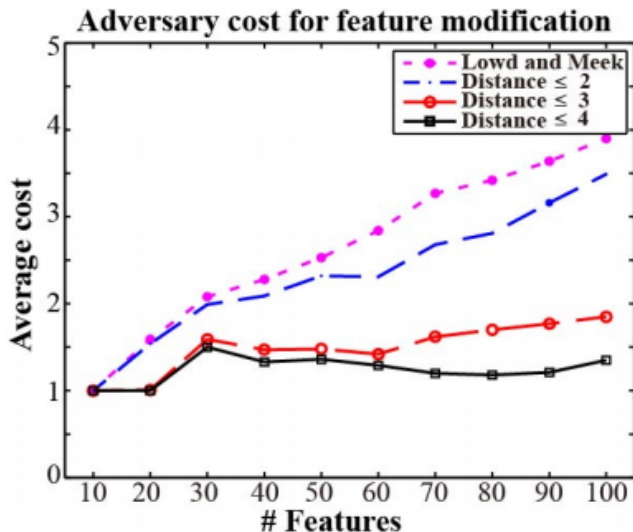
$$c(x', x^A) = \sum_i a_i |x'_i - x_i^A|,$$

- Equivalence-based cost functions

$$c(x', x^A) = \sum_i \min_{j \in F_i | x_j^A \oplus x'_j = 1} a_i |x'_j - x_i^A|,$$

Equivalence-based Cost Functions

- Better estimates the true cost of adversarial evasions



Modeling adversary's evasion attack

- Classifier $g(x)$ trained from data $\mathbf{x} \in \mathcal{D}$ and labels $\mathbf{y} \in \{+1, -1\}$
- Attacker's ideal malicious instance is x^A , but $g(x^A)$ likely close to 1
- Thus aims to optimize for x' :

$$\min_{x' \in X: g(x') = -1} c(x', x^A).$$

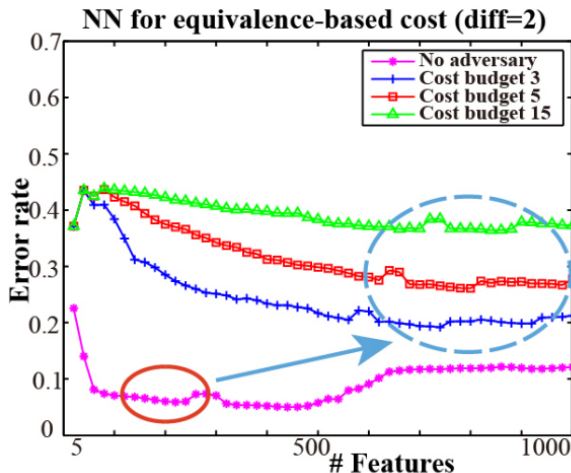
Too many features

- Suffers from the curse of dimensionality
- Standard practice: feature reduction
- Opens the door for adversarial evasion



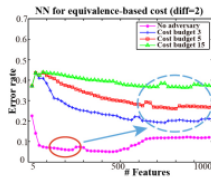
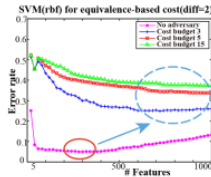
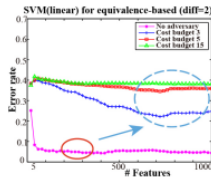
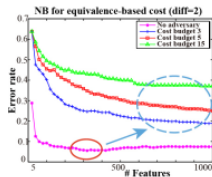
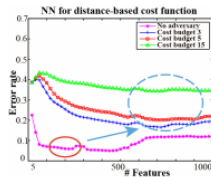
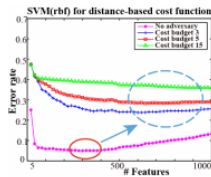
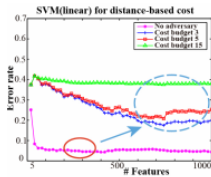
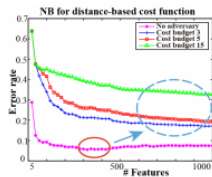
The Perils of Feature Reduction in Adversarial Classification

- Introduce Vulnerability to Evasions



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- Introduce Vulnerability to Evasions



(a)

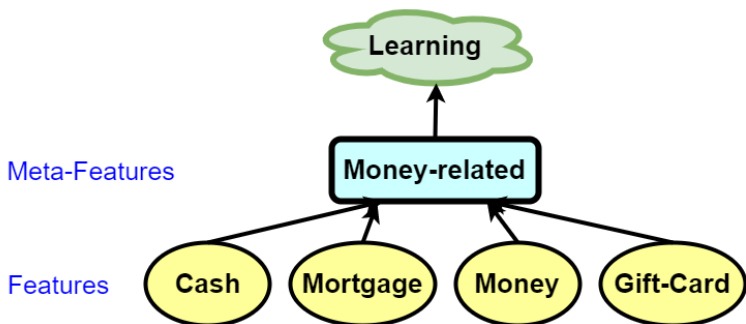
(b)

(c)

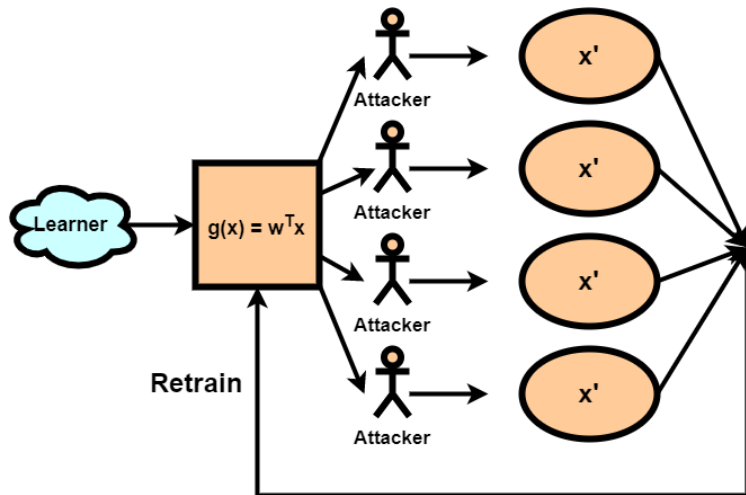
(d)

Solution 1: Equivalence-Based Classification

- A highly heuristic solution



Solution 2: Stackelberg Game Multi-Adversary Model



Modeling the problem

- optimization problem for the learner

$$\min_w \alpha \sum_{j|y_j=-1} l(-w^T x_j) + (1 - \alpha) \sum_{j|y_j=1} l(w^T F(x_j; w)) + \lambda \|w\|_1,$$

- $F(x_j; w) = z_i$ if $c(z_i, x_i) < B_i$ else x_i
- $z_i = \operatorname{argmin}_{x|w^T x \leq 0} c(x, x_i)$

Introduce Auxiliary Matrices to linearize as MILP

$$\min_{w,z,r} \alpha \sum_{i|y_i=0} D_i + (1-\alpha) \sum_{i|y_i=1} S_i + \lambda \sum_j K_j$$

$$\text{s.t. : } \forall a, i, j : z_i(a), r(a) \in \{0, 1\}$$

$$\sum_a z_i(a) = 1$$

$$\forall i : e_i = \sum_a m_i(a)(L_{ai}T_a + (1 - L_{ai})x_i)$$

$$\forall a, i, j : -Mz_i(a) \leq m_{ij}(a) \leq Mz_i(a)$$

$$\forall a, i, j : w_j - M(1 - z_i(a)) \leq m_{ij}(a) \leq w_j + M(1 - z_i(a))$$

$$\forall a : \sum_j w_j T_{aj} \leq 2 \sum_j T_{aj} y_{aj}$$

$$\forall a, j : -Mr_a \leq y_{aj} \leq Mr_a$$

$$\forall a, j : w_j - M(1 - r_a) \leq y_{aj} \leq w_j + M(1 - r_a)$$

$$\forall i : D_i = \max(0, 1 - w^T x_i)$$

$$\forall i : S_i = \max(0, 1 + e_i)$$

$$\forall j : K_j = \max(w_j, -w_j)$$

Algorithm 1 SMA(X)

$T = \text{randStrats}()$ // initial set of attacks

$X' \leftarrow \text{cluster}(X)$

$w_0 \leftarrow \text{MILP}(X', T)$

$w \leftarrow w_0$

while T changes **do**

for $x^A \in X'_{\text{spam}}$ **do**

$t = \text{computeAttack}(x^A, w)$

$T \leftarrow T \cup t$

end for

$w \leftarrow \text{MILP}(X', T)$

end while

return w

Still intractable

- Large number of ideal adversary instance x^A
- perform k-means clustering among malicious feature vector x^A
- Large number of constraints for each attack action $F(x_j, w)$
- Iteratively add constraint of best response to current w

Operational Decision

- Introduce random operation decision function $q(x, g(x))$ as a function of the learner



Reformulate the problem

- The attacker's utility is defined as:

$$\mu(x, x'; q) = V(x)Q(x, x')(1 - q(x')),$$

- where $Q(x, x') = e^{-\delta \|x - x'\|}$
- The defender's utility is then defined as:

$$U_{\mathcal{D}}(q, p) = \mathbb{E}_x [(1 - q(x))G(x)(1 - p(x)) - p(x)(P_S v_S(x; q) + P_A v_A(x; q))].$$

- which is equivalent to minimizing when $P_S = 1$:

$$\mathbb{E}_x [q(x)(1 - p(x)) + p(x)(1 - q(x))],$$

As a LP problem

- The learner's objective can be maximized using the following LP problem
- $q(x)$ must be defined over the entire feature space $\mathcal{X}$

$$\max_q \quad \hat{U}_{\mathcal{D}}(q, p) \quad (3a)$$

$$\text{s.t. : } 0 \leq q(x) \leq 1 \quad \forall x \in \mathcal{X} \quad (3b)$$

$$v_A(x; q) \geq \mu(x, x'; q) \quad \forall x, x' \in \mathcal{X} \quad (3c)$$

$$v_S(x; q) = V(x)(1 - q(x)) \quad \forall x \in \mathcal{X} \quad (3d)$$

$$\mathbb{E}_x[q(x)] \leq c, \quad (3e)$$

Formulate $q(x)$ using a set of normalised basis functions

$$q(x) = \sum_j \alpha_j \phi_j(x)$$

- The modified LP problem is tractable when the set of basis is not too large

$$\min_{\alpha \geq 0} \sum_j \alpha_j \mathbb{E}[G(x)\phi_j(x)(1 - p(x))] + \mathbb{E}[V(x)p(x)Q(x, \alpha)] \quad (4a)$$

$$\text{s.t. : } Q(x, \alpha) \geq e^{-\delta \|x - x'\|} (1 - \sum_i \alpha_i \phi_i(x')) \quad \forall x, x' \in \mathcal{X} \quad (4b)$$

$$\sum_j \alpha_j \mathbb{E}[\phi_j(x)] \leq c \quad (4c)$$

$$\sum_j \alpha_j \leq 1. \quad (4d)$$

Algorithm 1 AAS(X)

```
 $\phi = \text{ConstructBasis}()$   
 $\bar{\mathcal{X}} \leftarrow X$   
 $q \leftarrow \text{MASTER}(\bar{\mathcal{X}})$   
while true do  
  for  $x \in X_{bad}$  do  
     $x' = \mathcal{O}(x; q)$   
     $\bar{\mathcal{X}} \leftarrow \bar{\mathcal{X}} \cup x'$   
  end for  
  if All  $x' \in \bar{\mathcal{X}}$  then  
    // If no new  $x'$  generated  
    return  $q$   
  end if  
   $q \leftarrow \text{MASTER}(\bar{\mathcal{X}})$   
end while
```

Thanks For Watching